

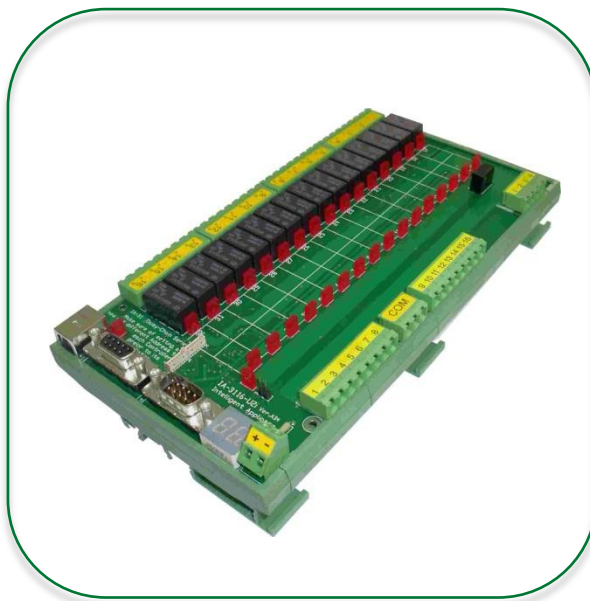
User Manual

IA-3116-U2i

16-ch Relay

16-ch Isolated Digital Inputs

USB or RS-232 Controlled



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www.intelligent-appliance.com

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We at Intelligent-Appliance highly value your opinion. Please feel free to contact us with your impression on any subject, or with any question or comment you may have.

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Introduction

The IA-3116-U2i is an industrial Relay controller with 16-ch Isolated Digital Input, USB and RS-232 controlled. The IA-3116U2i includes Isolated USB port, for better isolation between the host computer to the Factory floor, 16-ch 2Amp Relay, 16-ch Isolated Digital Input with a dual operation method and transparent chaining capabilities. The IA-3116-U2i includes a dual watchdog protection circuit, powered by 24VDC or 12VDC source, and includes a DIN-Rail mounting fixture. The IA-3116U2i software package includes Setting and Operation utilities, as well as open code software sample for fast system integration.

Features

- Isolated USB Port
- 16-ch SPST / SPDT Relays
- 16-ch Isolated Digital Input
- Wide Digital Input Range
- “Dry-contact” support
- GND and Positive Input Signals
- Onboard Dual Watchdog Protection
- Open Source code samples
- Setting and Operation Utility
- Simple Transparent Chaining
- DIN-Rail mounting ready

Specifications

Relays

Channels	16
Contact current	3Amp @ 30VDC 3Amp @ 250VAC
Contact method	SPDT, Form C, 4-ch SPST, Form A, 12-ch

Isolated Digital Inputs

Channels	16
Operation Methods	“Dry Contact” NPN and PNP source GND or positive signal
Voltage Range	TTL or up to 30VDC

Communication

Main COM	Isolated USB or RS-232, DB9 Female		
Expansion Port	Buffered RS-232, DB9 Male		
Factory Defaults	Baud Rate	19.2bps	
	Data bits	8	
	Parity	None	
	Stop bits	1	
Baud Rate Range	1200-115Kbps		
Host USB Cable	USB A/B, Included		
Host RS-232 Cable	DB9 M/F Straight Through		
Expansion RS-232 Cable	DB9 M/F Straight Through		

General

Power supply	24VDC, 0.4Amp
	12VDC, 0.8Amp (IA-3116-XXX-X-12VDC Version)
Module Size	265x115x45 mm
Mounting	Din-Rail Ready
Weight	450 gr

Ordering Information

- IA-3116-U2i-P:
16-ch Relay, 16-ch Isolated Digital Inputs
USB or RS-232 Controlled, 24VDC Powered
Pluggable Terminal Blocks Version
- IA-3116-E-P:
16-ch Relay, 16-ch Isolated Digital Inputs
Ethernet Controlled, 24VDC Powered
Pluggable Terminal Blocks Version



Warning & Safety

Intelligent Appliance products are NOT authorized for use as components in life support devices or systems.

Do not operate the device in a manner not specified in the documentation. Misuse of the device may result in injury and/or damage equipment.

When wiring the device disconnect it from the power source and turn OFF all connected devices.

Not doing so may result in electric shock, injury and/or damage your equipment.

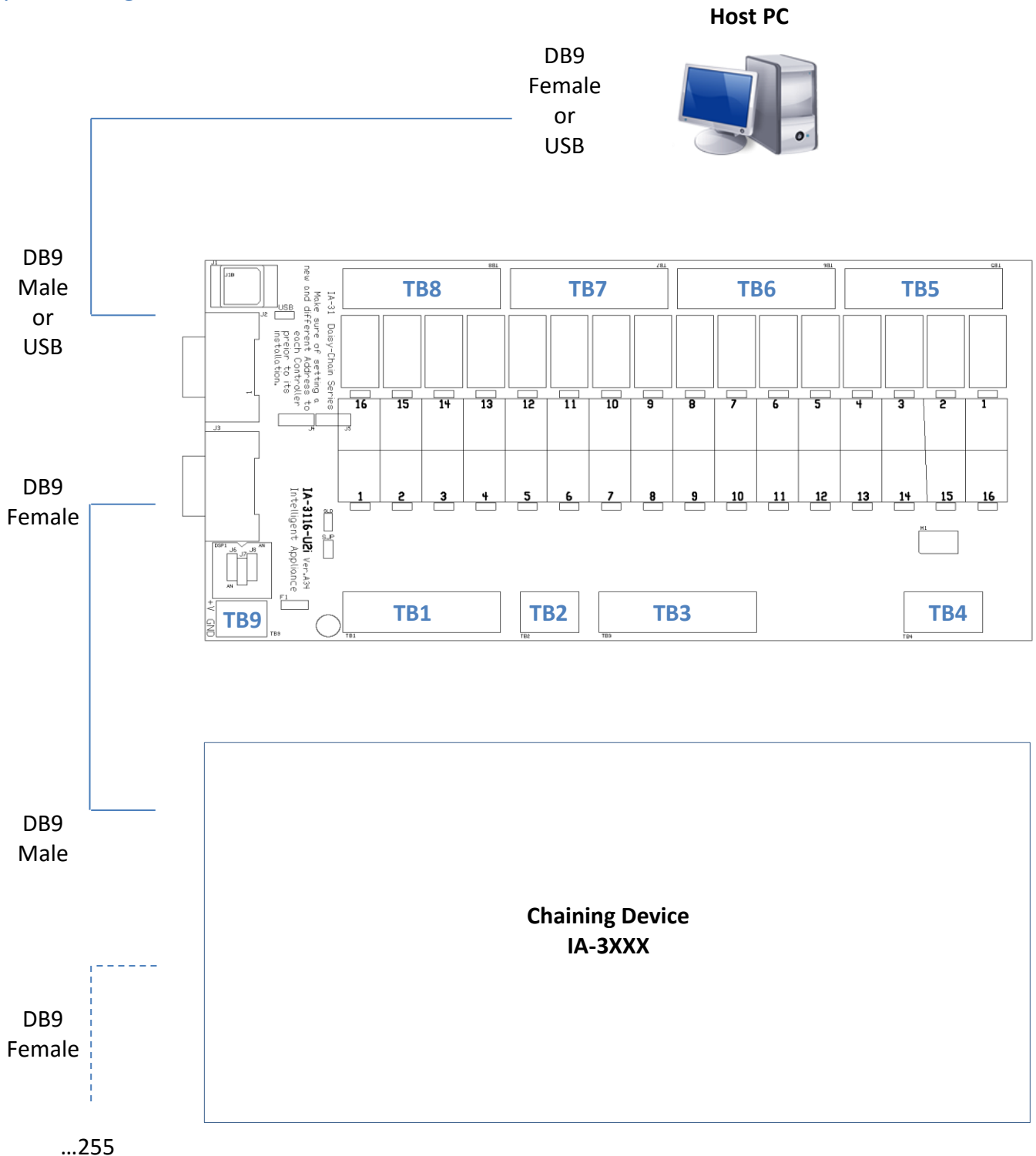
IA-3116-U2i

16-ch Relay
16-ch Isolated Digital Inputs
USB or RS-232 Controlled

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Installation

System Wiring



Pin Assignment

J1 – Main Port (USB)

J2 - Secondary Port (RS-232, DB9 Female)

Pin	Function
#2	Tx
#3	Rx
#5	GND

J3 – Expansion Port (RS-232, DB9 Male)

Pin	Function
#2	Rx
#3	Tx
#5	GND

TB1, TB3 – Isolated Digital input signals

TB2 – Isolated Digital input common

TB4 – Relay #1 to Relay #4 – NC contact

TB5– Relay #1 to Relay #4 – C and NO contacts

TB6 to TB8 – Relay #5 to Relay #16 – C and NO contacts

TB9 – Power supply input terminal

User Defined Jumpers & Led

SJP Jumper – User defined jumper

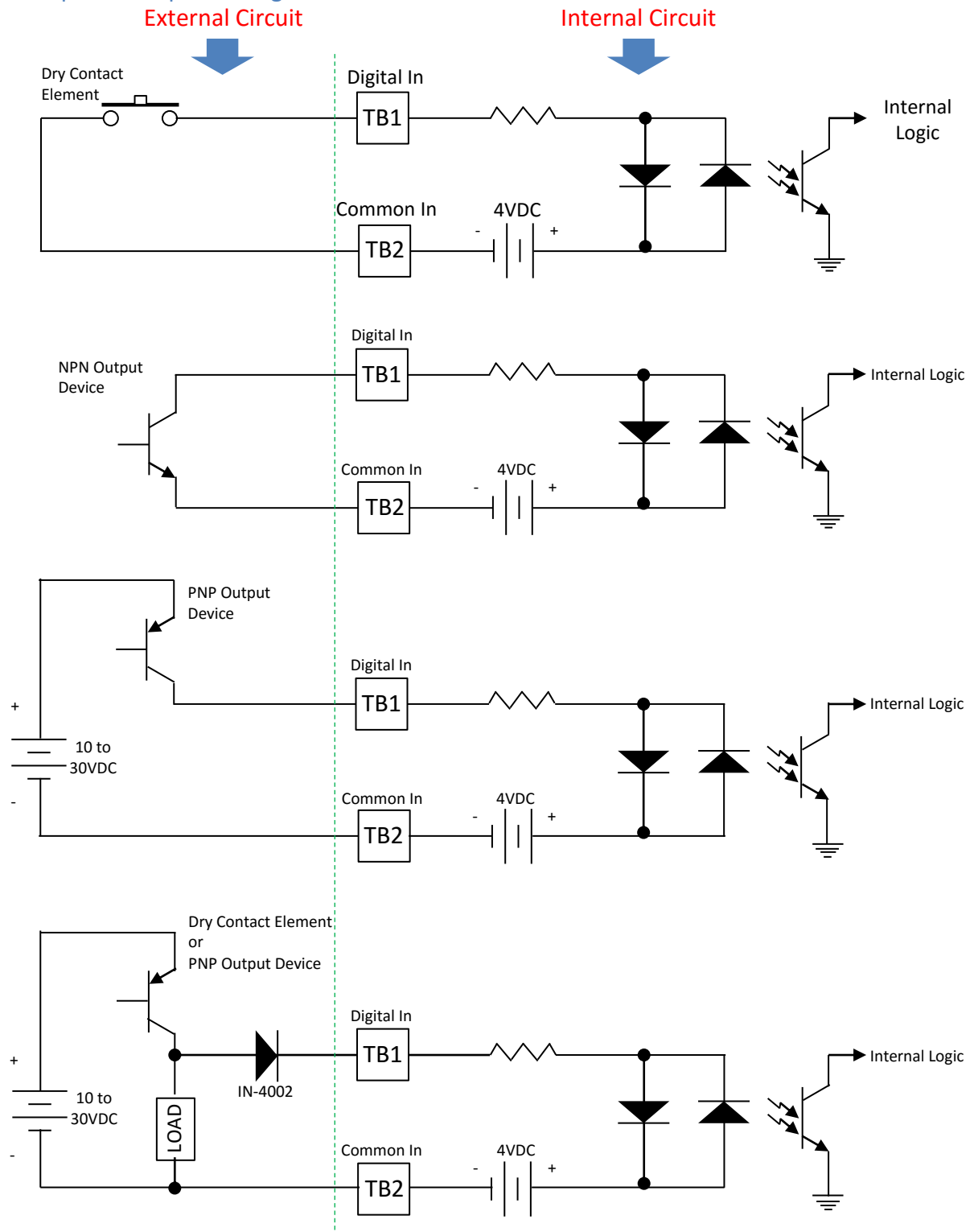
SLD Led – User defined led

Internal / External Power Source Isolators Jumpers

Internal – Dry contact, NPN, PNP support – Internally powered

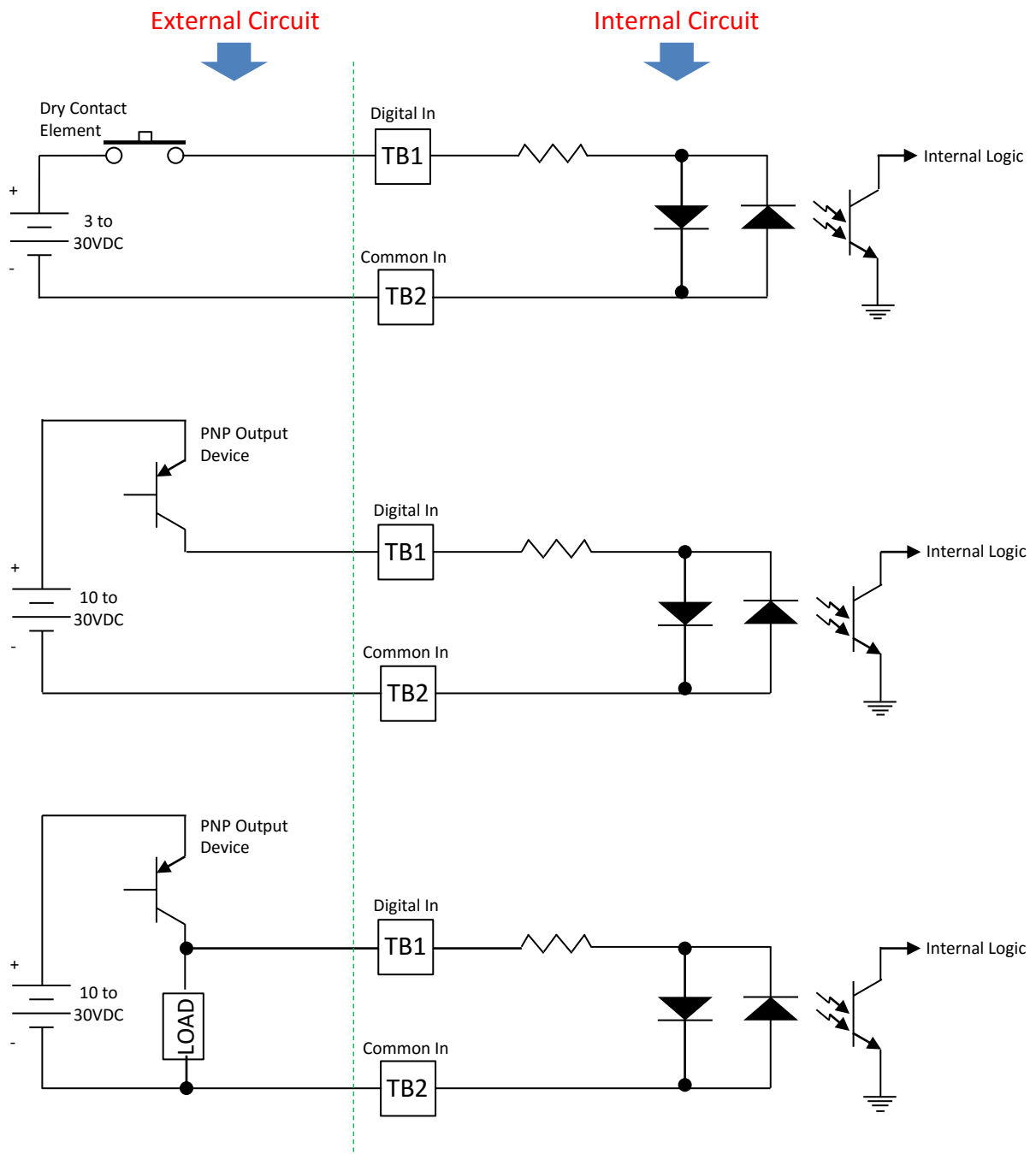
External – Loaded and special cases – Externally powered

Isolated Inputs - Jumper Settings: Internal Power



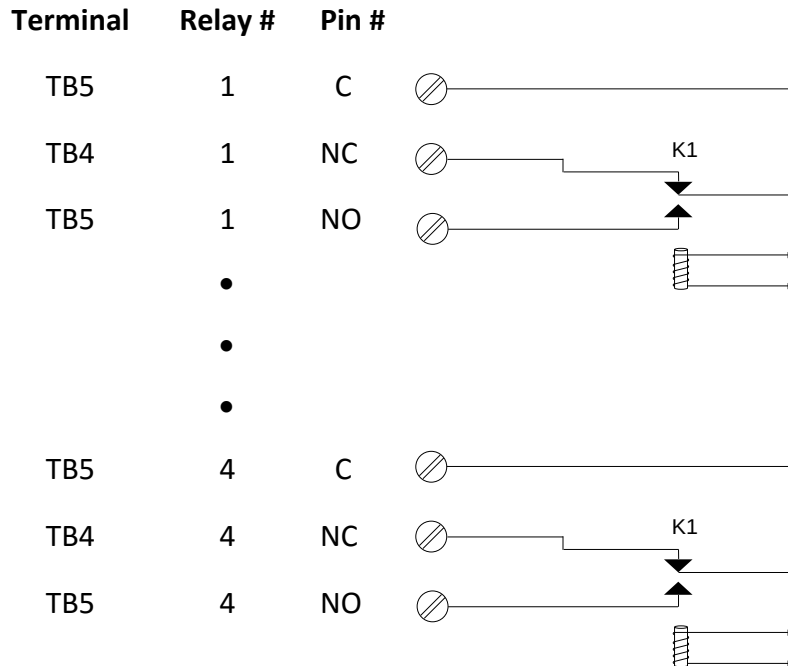
In case of Parallel loading, an external parallel diode should be added or “External Power” setting should be chosen.

Isolated Inputs - Jumper Settings: External Power

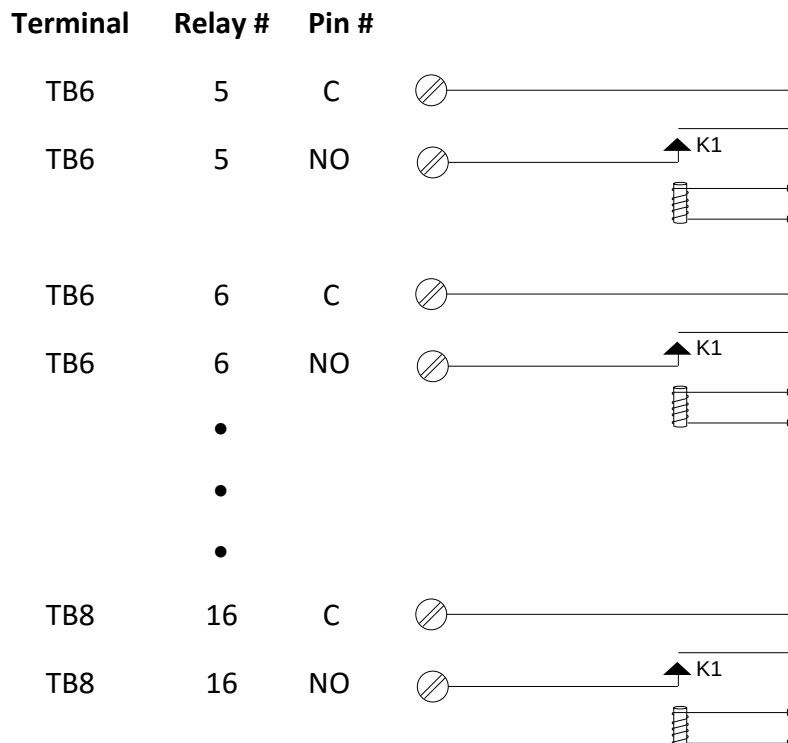


Relay Layout

SPDT, Form C, Channels 1-4



SPST, Form A, Channels 5-16



Software Installation

USB Port setup

Connect USB A/B Cable between the IA-3116-U2i to the host computer.

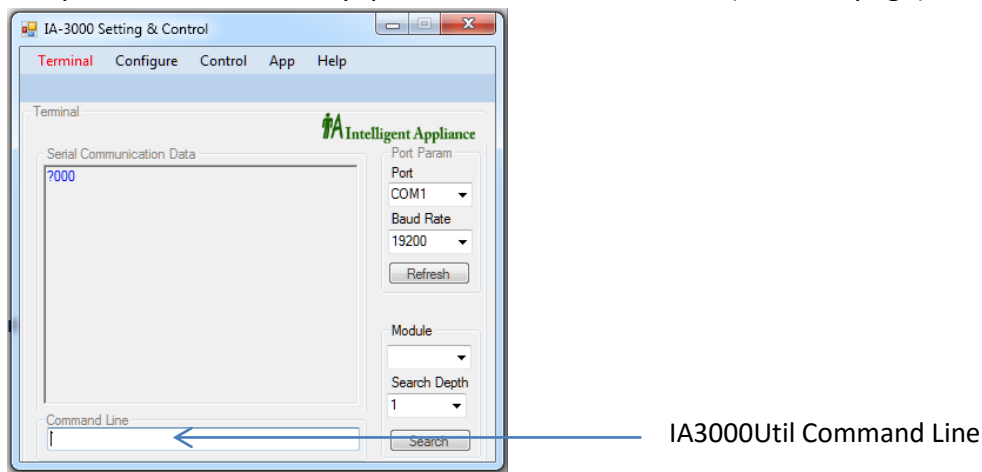
The computer informs on locating a new USB device, and asks for S/W drivers.

Kindly choose the USB-Drivers directory on the IA-3000 CD or from our website:

www.intelligent-appliance.com, and complete the task by pressing 'Next' and 'Finish' while asked for.

Job done will be accomplished by a steady lighting of the USB led on the IA-3116-U2i unit, and by a creation of new Serial COM that can be easily found on the Device Manager screen.

At this stage you can easily control the IA-3116-U2i I/O's by either any serial control software, or by the IA3000Util Utility, provided in the IA-3000 CD (see next page).



Locating the new COM port

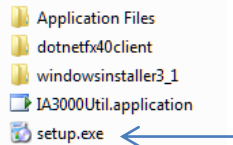
Start the 'Device Manager' utility. (Usually by selecting 'My Computer', Right Clicking the mouse button, choosing manage, Left Clicking and then double Left clicking on the Device Manager will list hardware items).

Select the '+' character to the left of the 'Ports (COM&LPT)', and you'll get a line that will define for example: 'USB Serial Port (COM4)'.

This line informs us that we should refer to COM4, in this case, in order to control the IA-3116-U2i while connected to this computer through its USB port.

IA-3000 Utility

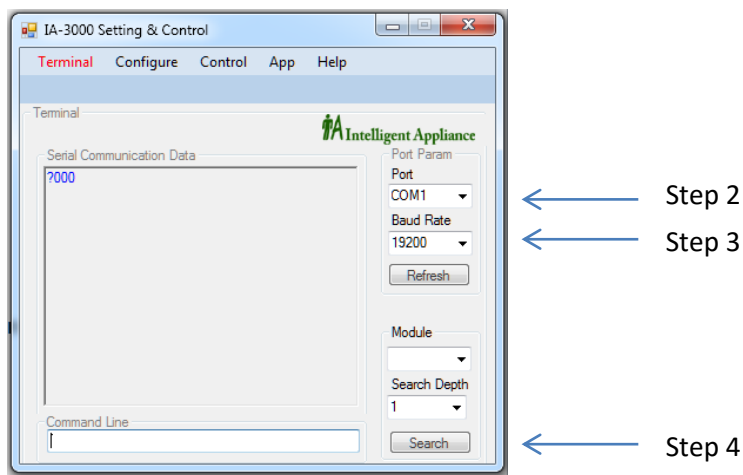
Install the IA3000Util Utility in your computer by clicking on the 'Setup' icon in the 'IA-Utility' directory, on the IA-3000 CD.



Or download it from our online Knowledge Base. (www.intelligent-appliance.com)

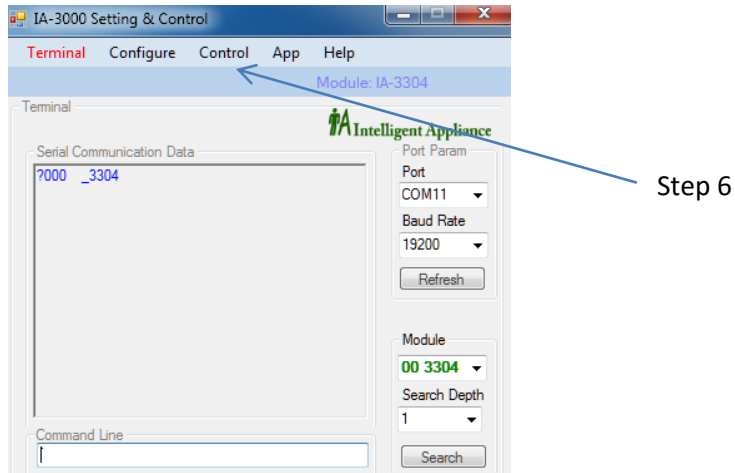
Handling IA-3000 Utility

1. Start the IA3000Util Utility by pressing 'Start Manu' on the computer's main screen, select 'All Programs', and finally 'IA3000Util'.
2. Select the appropriate COM



3. Select '19200' to fit into the right of the 'Baud' label (in case that the IA-3116-U2i is at its default setting stage).
4. Press the 'Search' button and wait for the utility to list all chained items.
5. Select the desired device out of the Module list that appears above the 'Search' button.

6. Once the device is selected, its form will be shown on the control panel label.



7. Left clicking the buttons will activate or dis-activate the appropriate I/O.

Command Set

The following table is a quick reference table for the IA-3116-U2i, A host computer / PLC may control the IA-3116-U2i by simply sending ASCII commands though a standard COM port. Each command is structured from a delimiter character, modules address, command character, data if any carriage returns character. All commands must use UPPER CASE characters.

?AA0 – GET DEVICE NAME	16
?AA1 – GET DEVICE FIRMWARE VERSION	17
?AA2 – GET INPUTS AND OUTPUTS STATUS.....	18
?AA50 – GET DEVICE MODE (REGISTER #50).....	19
?AA51 – GET DEVICE MODE (REGISTER #51).....	20
?AAID – GET MODULE’S ID NUMBER	21
?AAS – GET JUMPER & LED STATUS	22
?AAWDT – GET WD2 COUNT-DOWN VALUE	23
!AA2DDDD – SET RELAY STATUS	24
!AA3DD – ACTIVATE RELAY N	25
!AA4DD – DE ACTIVATE RELAY N	26
!AA50DD – SET DEVICE MODE REGISTER #50 VALUE.....	27
!AA51DD – SET DEVICE MODE REGISTER #51 VALUE.....	28
!AA6DD – SET BAUD RATE.....	29
!AA7DD – SET MODULE’S ADDRESS.....	30
!AAEDDDD – SET INITIAL STATE	31
!AASDD – SET USER DEFINED LED	32
!AAWDTDD – SET WD2 TIME VALUE.....	33
^^E – FORCE INITIAL STATE AT ALL MODULES.....	34

?aa0 – Get device name

Description Request the Device model name. Can be used to identify the connected module type at the specified address.

Syntax ?aa0<CR>
? Delimiter character
aa Hexadecimal address of the device
0 Get device Model command
<CR> Carriage Return - End of command

Response _nnnn<CR> if the command was valid
_ Response delimiter
nnnn A string containing the device name
<CR> Carriage Return - end of response

Example Command: ?000<CR>
Response: _3116<CR>

Request the device at address 00Hex to send its model name.
The response indicates that the command was successful and that the device at this address is IA-3116-U2i

?aa1 – Get device firmware version

Description Request the Device version

Syntax ?aa1<CR>
? Delimiter character
aa Hexadecimal address of the device
1 Get device Version command
<CR> Carriage Return - End of command

Response _nnnn<CR> if the command was valid
_ Response delimiter
nnnn A string containing the device version
<CR> Carriage Return - end of response

Example Command: ?001<CR>
Response: _u157<CR>

Request the device at address 00 Hex to send its version.
The response indicates that the command was successful and that the device version at this address is u1.57

?aa2 – Get Inputs and Outputs status

Description Read digital inputs and relays present status

Syntax ?aa2<CR>
 ? Delimiter character
 aa Hexadecimal address of the device
 2 Read I/O status
 <CR> Carriage Return - End of command

Response _ABCDEFGH<CR> if the command was valid

A 1st Input nibble
 B 2nd Input nibble
 C 3rd Input nibble
 D 4th Input nibble
 E 1st Output nibble
 F 2nd Output nibble
 G 3rd Output nibble
 H 4th Output nibble

Input

3	2	1	0	3	2	1	0	3	2	1	0	3	2	1	0
D16	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1
A				B				C				D			

Output

3	2	1	0	3	2	1	0	3	2	1	0	3	2	1	0
R16	R15	R14	R13	R12	R11	R10	R9	R8	R7	R6	R5	R4	R3	R2	R1
E				F				G				H			

Example Command: ?002<CR>
 Response: _00018001<CR>

Input #1 is Active and both relays #1 and #16 are activated

?aa50 – Get Device mode (Register #50)

Description This command reads the module operation mode

Syntax ?aa50<CR>
? Delimiter character
aa Hexadecimal address of the device
50 System Mode command
<CR> Carriage Return - End of command

Response _dd<CR> if the command was valid
dd Mode (00-FF)

82	Enable BR change	Bit 7 & Bit 1
40	Eliminate Response	Bit 6
20	Add device Address to response	Bit 5
10	Bit level Report	Bit 4
04	Board level Report	Bit 2
02	Report on command errors	Bit 1
00	Normal	

<CR> Carriage Return - End of command

Example Command: ?0050<CR>
Response: _82<CR>

In this example the module operation mode enables baud rate change.
It will also send error messages for invalid commands.

?aa51 – Get Device mode (Register #51)

Description This command reads the device mode register #51 data.

Syntax ?aa51<CR>
? Delimiter character
aa Hexadecimal address of the device
51 Mode register #51 command
<CR> Carriage Return - End of command

bit	Value (dd)	Function	
		WD2 - Host communication watchdog:	
2	04	1	Enabled
	00	0	Disabled

Response _dd<CR> if the command was valid
dd Mode register #51

<CR> Carriage Return - End of command

Example Command: ?0051<CR>
Response: _04<CR>

In this example watchdog 2 is enabled.
This host communication protection watchdog, will force all relays to their WD2 special Relay Pattern state.
In case that no communication will take place to this card for over the time defined in command [!aaWDTdd](#)<CR>

?aaID – Get module's ID number

Description	This command reads the Device ID
Syntax	<pre>?aaID<CR></pre> ? Delimiter character aa Hexadecimal address of the device ID Command for read ID <CR> Carriage Return - End of command
Response	_ID nnnnnnnn
Example	Command: ?00ID<CR> Response: _ID 00412534<CR>
	In this example we read S/N of device #00

?aaS – Get jumper & led status

Description This command reads the status of JP1 and the LED.

Syntax ?aaS<CR>
 ? Delimiter character
 aa Hexadecimal address of the device
 S Read jumper & led status
 <CR> Carriage Return - End of command

Response _dd<CR> if the command was valid
 _ Delimiter character
 dd Output digits
 A 1st Input nibble
 B 2nd Input nibble

0	0	0	SJP	0	0	0	SLD
D7	D6	D5	D4	D3	D2	D1	D0
A				B			

SLD=1 LED is ON
 SLD=2 SJP jumper is CLOSED
 Command: ?00S<CR>
 Response: _11<CR>

Example

In this example the LED is ON and JP1 is CLOSE

?aaWDT – Get WD2 Count-down value

Description This command reads the System Communication Watchdog, WD2, Counter Count-down value and sets its value to the Preset value.

Syntax ?aaWDT<CR>
? Delimiter character
aa Hexadecimal address of the device
WDT Read jumper & led status
<CR> Carriage Return - End of command

Response _dd<CR> if the command was valid
_ Delimiter character
dd Output digits in HEX format

Example Command: ?00WDT<CR>

Response: _12<CR>

In case that WD2 is Activated!

Response: _20 WD2 ERR<CR>

In case that WD2 is Not Activated!

In this example WD2 Count-down value is 18 Seconds.

NOTE 1: This command reads the WD2 Counter value. Not the WDT Preset value that is defined by the “[!aaWDTdd<CR>](#)” command.

NOTE 2: In case that WD2 is not operated, this command will yield an error and it will show WDT predefined value.

NOTE 3: This command, in addition to reading WD2 Counter value, sets the WD2 Counter value to its Preset value.

NOTE 4: Each reading might yield a different value as it is depended on the time elapse since previous reading and presetting.

!aa2dddd – Set relay status

Description This command defines module's output state.

Syntax !aa2dddd <CR>
! Delimiter character
aa Hexadecimal address of the device
2 System control command
d Relay output activation command data for each nibble in hex format
<CR> Carriage Return - End of command

Response |dddd
if the command was valid and if FB messages are enabled

Example 1 (Normal)	Command: !0021111<CR> Response: 1111<CR> This command will activate relay #1, #5, #9, #13
Example 2 (REG 50, bit 5 Set) !aa5020	Command: !0021111<CR> Response: 001111<CR> This command will activate relay #1, #5, #9, #13 And includes the device address 00
Example 3 (REG 50, bit 6 Set) !aa5040	Command: !0021111<CR> Response: null This command will activate relay #1, #5, #9, #13 And no response will be received

!aa3dd – Activate relay N

Description This command activates a single relay.

Syntax !aa3dd <CR>
! Delimiter character
aa Hexadecimal address of the device
3 Single relay activation command
dd N Relay ID in hex format
<CR> Carriage Return - End of command

Response |Sdd if the command was valid

Example Command: !00302<CR>
Response: |S02<CR>

This command will activate relay #3 only (!) all other relays will be not changed.

Relay	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
DATA (dd)	0F	0E	0D	0C	0B	0A	09	08	07	06	05	04	03	02	01	00

!aa4dd – De activate relay N

Description This command De activates a single relay.

Syntax !aa4dd <CR>
! Delimiter character
aa Hexadecimal address of the device
4 De activate relay N command
dd N Relay ID in hex format
<CR> Carriage Return - End of command

Response |Cdd if the command was valid

Example Command: !00402<CR>
Response: |C02<CR>

This command will De activate relay #3 (!) all other relays state will not be changed.

Relay	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
DATA (dd)	0F	0E	0D	0C	0B	0A	09	08	07	06	05	04	03	02	01	00

!aa50dd – Set device mode Register #50 value

Description This command sets the power-up mode, enables/disables error messages, Address, Baud Rate, Register #51 values change, Bit Level and Digital Input bit level change report.

Syntax !aa50dd <CR>
!
aa
50
dd

Delimiter character
Hexadecimal address of the device
System mode command
Mode (00-FF)

82	Enable BR change	Bit 7 & Bit 1
40	Eliminate Response	Bit 6 *
20	Add device Address to response	Bit 5 *
10	Bit level Report	Bit 4
04	Board level Report	Bit 2
02	Report on command errors	Bit 1
00	Normal	

<CR> Carriage Return - End of command

Response |dd EE OK if the command was valid

Example Command: !005082<CR>
Response: |82 EE OK

This command will enable Address , BR changing and enable the device error messages.
(Error messages are sent in response to invalid commands) and will disable Address and baud rate changed by mistake.
Make sure setting a normal mode like '00' or '02' right after changing the BR to disable mistakenly done BR change.

NOTE! * See command [!aa2ddddd<CR>](#) for more details about bits 5 & 6

!aa51dd – Set device mode Register #51 value

Description This command enables/disables WD2.

Note

- One must set the device [mode register #50 value to '82'](#) to enable setting of device mode register #51 value.

Syntax !aa51dd <CR>

! Delimiter character
aa Hexadecimal address of the device
51 System mode command
dd Mode register #51

bit	Value (dd)	Function		
		WD2 - Host communication watchdog:		
2	04		1	Enabled
	00		0	Disabled

<CR> Carriage Return - End of command

Response |dd EE OK if the command was valid

Example Command: !005104<CR>
Response: |04 EE OK

This command will activate System Communication Watchdog WD2

NOTE!

- Activating WD2 should be accomplished by polling the device repeatedly in a time period that will not exceed the [WD2 time set](#).
- Exceeding this value will set all relays to WD2 special Relay Pattern, '8000': Relay #32 is activated, all other relays are deactivated.

!aa6dd – Set baud rate

Description For compatibility with existing devices the IA-3116-U2i can be set to other standard baud rates

Syntax !aa6dd <CR>
! Delimiter character
aa Hexadecimal address of the device
6 Change device baud rate command
dd Two characters representing the desired baud rate:
12 1200
24 2400
48 4800
96 9600
19 19200 (default)
38 38400
57 57600
11 115200
<CR> Carriage Return - End of command

Response |dd<CR> if the command was valid
| Response delimiter
dd New baud rate
<CR> Carriage Return - End of response

Example Command: !01696<CR>
Response: |96<CR>

Change the baud rate of the device at address 01Hex to 9600



1. Mode must be set to "82" first. ([!005082](#))
2. Changes will take effect after the next power up. (Power off)
3. Make sure setting the mode back to normal right after changing the BR.

!aa7dd – Set module's address

Description Each device must have a unique network address.
This command defines the module's address.
[Mode REG #50](#) must be set to '82' in order to make use of this command

Syntax !aa7dd <CR>
! Delimiter character
aa Hexadecimal address of the device
7 Change device baud rate command
dd New Hexadecimal address
<CR> Carriage Return - End of command

Response |dd<CR> if the command was valid

Example Command: !00701<CR>
Response: |01<CR>

Change the address of the device at address 0(Hex) to 1(Hex)



1. Mode REG 50 must be set to '82' in order to make use of this command [!aa5082](#)<CR>
2. Factory default is 00Hex
3. In products chained system, each product must be set to a unique address.
4. The updated address is displayed on the boards 7 segment led display.

!aaEdddd – Set initial state

Description This command defines relays state at POWER-UP

Syntax !aaEdddd <CR>
! Delimiter character
aa Hexadecimal address of the device
E System control command
d Relay output activation command data for each nibble in hex format
<CR> Carriage Return - End of command

Response |Edddd<CR> if the command was valid

Example Command: !00E1234<CR>
Response: |E 1234 EE OK<CR>

This command will define the initiate state of all relays and will activate it for user's verification.



Note

Factory Default state is !00E0000 (All relays are OFF at power up)

!aaSdd – Set user defined led

Description This command turns ON or turns OFF the SLD LED

Syntax !aaSdd <CR>
! Delimiter character
aa Hexadecimal address of the device
S Set Led status
0d LED status:
01 On
00 Off
<CR> Carriage Return - End of command

Response |0d<CR> if the command was valid

Example Command: !00S01<CR>
Response: |01<CR>

This command will turn on the LED



Note

The default LED status is ON.

!aaWDTdd – Set WD2 Time value.

Description This command Set WD2 Time value

Syntax !aaWDTdd<CR>
! Delimiter character
aa Hexadecimal address of the device
WDT Set WD2 Relays Status
dd 2 Hex characters defines 10 to 256 Seconds.

<CR> Carriage Return - End of command

Response |dd<CR> if the command was valid

Example Command: !00WDT20<CR>
Response: |20<CR>

This command will activate WD2 relays pattern within 32 Seconds.



Note

- The default WD2 time set is “20”. WD2 is NOT Active.
- It is recommended to set each chained device to a different value in order to receive its message.
- Exceeding this value will set all relays to WD2 special Relay Pattern, ‘8000’: Relay #16 is activated, all other relays are deactivated.

^^E – Force initial state at all modules

Description This command forces all modules on present chain into their pre-defined initiate state.

Syntax ^^E<CR>
^^ Delimiter character
E System control command
<CR> Carriage Return - End of command

Response No Response message

Example Command: ^^E<CR>
Response: No Response message

